

Beregningsteknik for Computer Ingeniører

MM 2: Sets and combinatorics

Topics:

Definition of a set and set operations

- Set notation (set-builder), Venn diagrams, identities

Combinatorics: principles of counting

- Permutations and number of combinations of r elements (out of n elements)
- Relation to binomial coefficients

Sets

- **Definition:** A *set* is an unordered collection of objects, called *elements* or *members* of the set. A set is said to *contain* its elements. We write $a \in A$ to denote that a is an element of the set A . The notation $a \notin A$ denotes that a is not an element of the set A .

- Examples ("universes") U:

$\mathbb{N} = \{0, 1, 2, 3, \dots\}$, the set of **natural numbers**
 $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$, the set of **integers**
 $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$, the set of **positive integers**
 $\mathbb{Q} = \{p/q \mid p \in \mathbb{Z}, q \in \mathbb{Z}, \text{ and } q \neq 0\}$, the set of **rational numbers**
 $\mathbb{R}$, the set of **real numbers**
 $\mathbb{R}^+$, the set of **positive real numbers**
 $\mathbb{C}$, the set of **complex numbers**.

- Set definition

$$S = \{1, 3, 5, 7, \dots\}$$

$$S = \{x \in \mathbb{Z}^+ \mid 2 \text{ divides } (x + 1)\}$$

Set-builder notation, stating properties
(a condition) that needs to be true

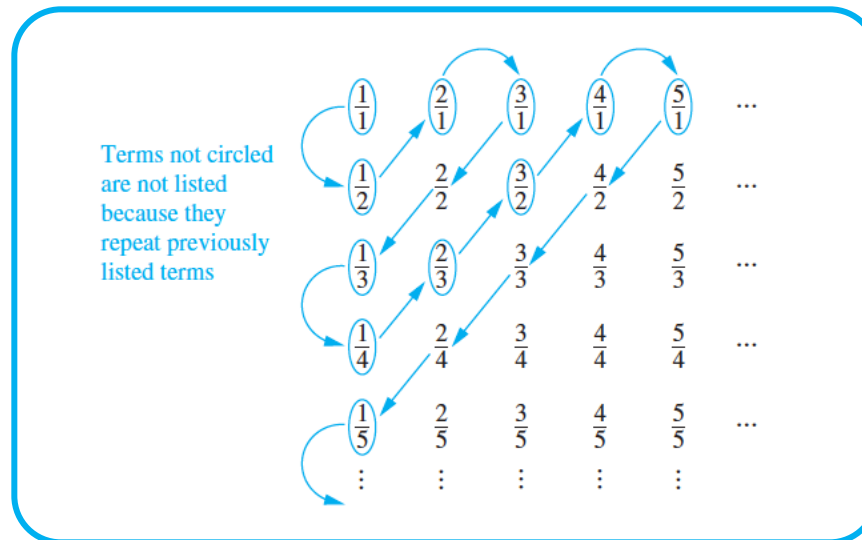
$$A = B \iff \forall x \in \mathbb{N} (x \in A \iff x \in B)$$

Set equality using a **quantification**
(that needs to be true)

Countable sets

- Definition:

A set that is either finite or has the same cardinality as the set of positive integers is called *countable*. A set that is not countable is called *uncountable*. When an infinite set S is countable, we denote the cardinality of S by $\aleph_0$ (where $\aleph$ is aleph, the first letter of the Hebrew alphabet). We write $|S| = \aleph_0$ and say that S has cardinality “aleph null.”
- Example: the positive rational numbers are countable



Set operations

- **Cardinality**: The number of elements in a set; $|\emptyset| = 0$
- **Union** of A and B:
 - $A \cup B$ {all elements that are either in A or B}
- **Intersection** of A and B:
 - $A \cap B$ {all elements that are both in A and B}
- Two sets are mutually **exclusive**, if $A \cap B = \emptyset$
- The **complement** of a set:
 - $\bar{A}$ {all elements that are not in A}
- If all elements of B are in A, B is **contained** in A:
 - B is a subset, $B \subseteq A$
 - B is a proper subset, $B \subset A$

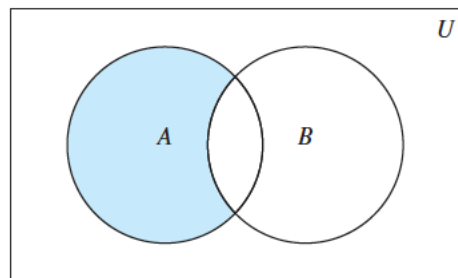
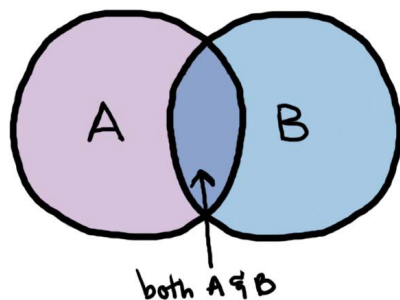
Set operations

- Difference:

Let A and B be sets. The *difference* of A and B , denoted by $A - B$, is the set containing those elements that are in A but not in B . The difference of A and B is also called the *complement of B with respect to A* .

Remark: The difference of sets A and B is sometimes denoted by $A \setminus B$.

VENN DIAGRAM!



$A - B$ is shaded.

- Example: with help of Venn diagram show that

$$A - B = A \cap \overline{B}$$

Set identities

- Set identities correspond to logical identities (laws):
 - identity, domination, idempotent, complement's complement, commutative, associative, distributive, De Morgan's, absorption, complement

notation: $E \cap F$

Commutative law $E \cup F = F \cup E$

$EF = FE$

Associative law $(E \cup F) \cup G = E \cup (F \cup G)$

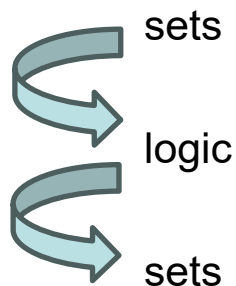
$(EF)G = E(FG)$

Distributive law $(E \cup F)G = EG \cup FG$

$EF \cup G = (E \cup G)(F \cup G)$

Set identities

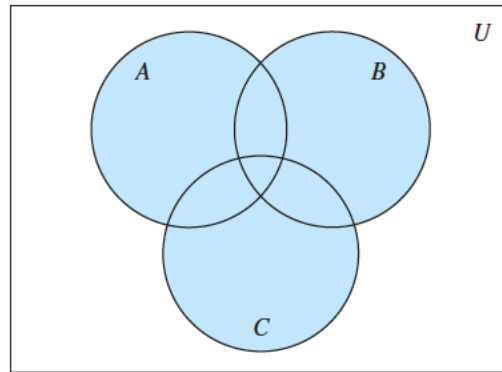
- Proving set identities
 - Show that they are subsets of one another
 - Show using a membership table
 - Show by applying identities
- Example: Proof of De Morgan's first law via set builder notation and logical equivalences:



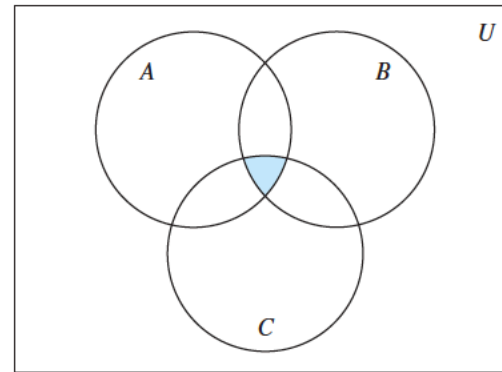
$$\begin{aligned}\overline{A \cap B} &= \{x \mid x \notin A \cap B\} && \text{by definition of complement} \\ &= \{x \mid \neg(x \in (A \cap B))\} && \text{by definition of does not belong symbol} \\ &= \{x \mid \neg(x \in A \wedge x \in B)\} && \text{by definition of intersection} \\ &= \{x \mid \neg(x \in A) \vee \neg(x \in B)\} && \text{by the first De Morgan law for logical equivalences} \\ &= \{x \mid x \notin A \vee x \notin B\} && \text{by definition of does not belong symbol} \\ &= \{x \mid x \in \overline{A} \vee x \in \overline{B}\} && \text{by definition of complement} \\ &= \{x \mid x \in \overline{A} \cup \overline{B}\} && \text{by definition of union} \\ &= \overline{A} \cup \overline{B} && \text{by meaning of set builder notation}\end{aligned}$$

Operations with n sets

- The definitions for set operations can be generalized for the case of n sets



(a) $A \cup B \cup C$ is shaded.



(b) $A \cap B \cap C$ is shaded.

- Notation used:

$$A_1 \cup A_2 \cup \dots \cup A_n = \bigcup_{i=1}^n A_i$$

$$A_1 \cap A_2 \cap \dots \cap A_n = \bigcap_{i=1}^n A_i$$

Combinatorics

- **Combinatorics** is an area of mathematics primarily concerned with **counting**
- **Principles of counting**
 - Suppose 2 experiments are to be performed. If there are m possible outcomes for experiment 1, and for each possible outcome of experiment 1, there are k possible outcomes for experiment 2, then there are mk possible outcomes of the 2 experiments
 - If a third experiment is to be performed, with l possible outcomes $\rightarrow$ sample space of 3 experiments consists of $mk l$ elements

Principles of counting

THE PRODUCT RULE Suppose that a procedure can be broken down into a sequence of two tasks. If there are n_1 ways to do the first task and for each of these ways of doing the first task, there are n_2 ways to do the second task, then there are $n_1 n_2$ ways to do the procedure.

THE SUM RULE If a task can be done either in one of n_1 ways or in one of n_2 ways, where none of the set of n_1 ways is the same as any of the set of n_2 ways, then there are $n_1 + n_2$ ways to do the task.

THE SUBTRACTION RULE If a task can be done in either n_1 ways or n_2 ways, then the number of ways to do the task is $n_1 + n_2$ minus the number of ways to do the task that are common to the two different ways.

$$|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$$

Combinatoric problems

- (wishful thinking) AAU is making 24 group rooms available for 3 COMTEK groups. Each group is getting one room, which leaves **X possible assignments**
- A university board position is to be filled by either a student or a faculty member. Out of 250 students and 50 faculty members there are **X possible selections**
- Out of all bit strings of length 8 ($b_7b_6b_5b_4b_3b_2b_1b_0$) there are **X which either start with 1 (b_7) or end in 00 (b_1b_0)**

Which X and how did you count it?

Permutations

- Definition: **Permutation** is an **ordered** sequence of objects
- An ordered arrangement of r objects from a set is called an **r -permutation, $P(n,r)$**

If n is a positive integer and r is an integer with $1 \leq r \leq n$, then there are

$$P(n, r) = n(n-1)(n-2) \cdots (n-r+1) = \frac{n!}{(n-r)!}$$

r -permutations of a set with n distinct elements.

Combinations

THE DIVISION RULE "There are $P(n,k) / P(k,k)$ ways to do a task if it can be done using a procedure that can be carried out in $P(n,k)$ ways, and for every way, exactly $P(k,k)$ ways correspond to the same way"

- Definition: **Combination** is an **unordered** sequence of objects
- An unordered arrangement of k objects from a set is called an **k -combination, $C(n,k)$**

$$C(n,k) = \frac{n(n-1) \cdots (n-k+1)}{k!} = \frac{n!}{(n-k)!k!} = \binom{n}{k}$$

Sampling with/without Replacement and with/without Ordering

n objects in the basket, we choose k objects.
Number of possible outcomes?

- Sampling with Replacement and with Ordering n^k
- Sampling without Replacement and with Ordering (**permutation**)
$$n(n-1) \cdots (n-k+1) = \frac{n!}{(n-k)!}$$
- Sampling without Replacement and without Ordering (**combination**)

$$\frac{n(n-1) \cdots (n-k+1)}{k!} = \frac{n!}{(n-k)!k!} = \binom{n}{k}$$

Binom.
Coeff.

- Sampling with Replacement and without Ordering $\binom{n-1+k}{k}$

Binomial coefficients

- The number of combinations of k elements from a set of n elements $\binom{n}{k}$ is also called binomial coefficients.

THE BINOMIAL THEOREM Let x and y be variables, and let n be a nonnegative integer. Then

$$(x + y)^n = \sum_{j=0}^n \binom{n}{j} x^{n-j} y^j = \binom{n}{0} x^n + \binom{n}{1} x^{n-1} y + \cdots + \binom{n}{n-1} x y^{n-1} + \binom{n}{n} y^n.$$

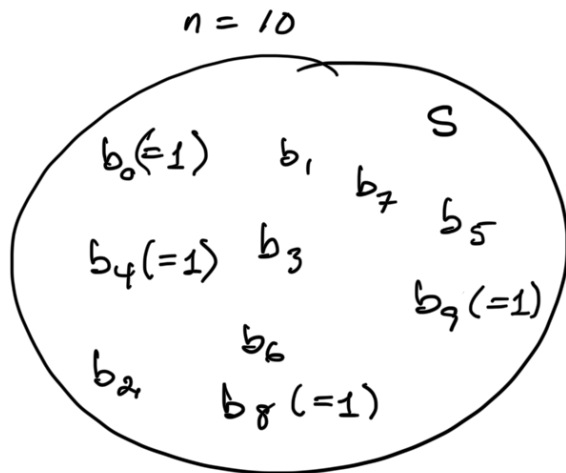
- Why are they called binomial?

$$\sum_{k=0}^n \binom{n}{k} = 2^n.$$

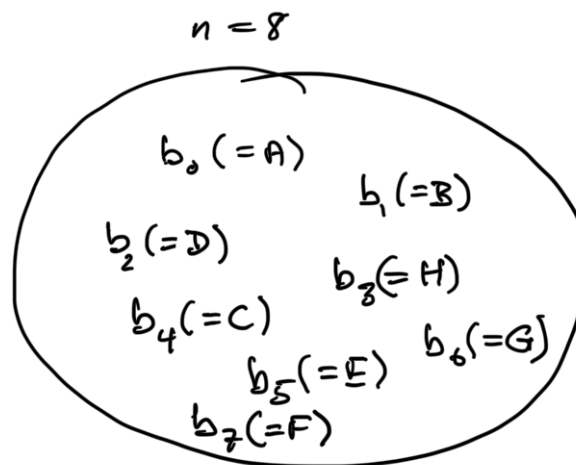
» "The number 2 to the power of n written as a sum of binomial coefficients"

Exercises and sampling

- Strings of bits example ($n = 10$) with exactly four 1's
- Assignment not done beforehand! (not instantiated)
- Individual elements are implicit



- Sequences of letters ($n = 8$) with permutations or combinations
- Assignment done beforehand! (instantiated)
- Individual elements are explicit



- Clearly, we can assign order, but not necessarily!
- Also, we can replace, but not necessarily!